

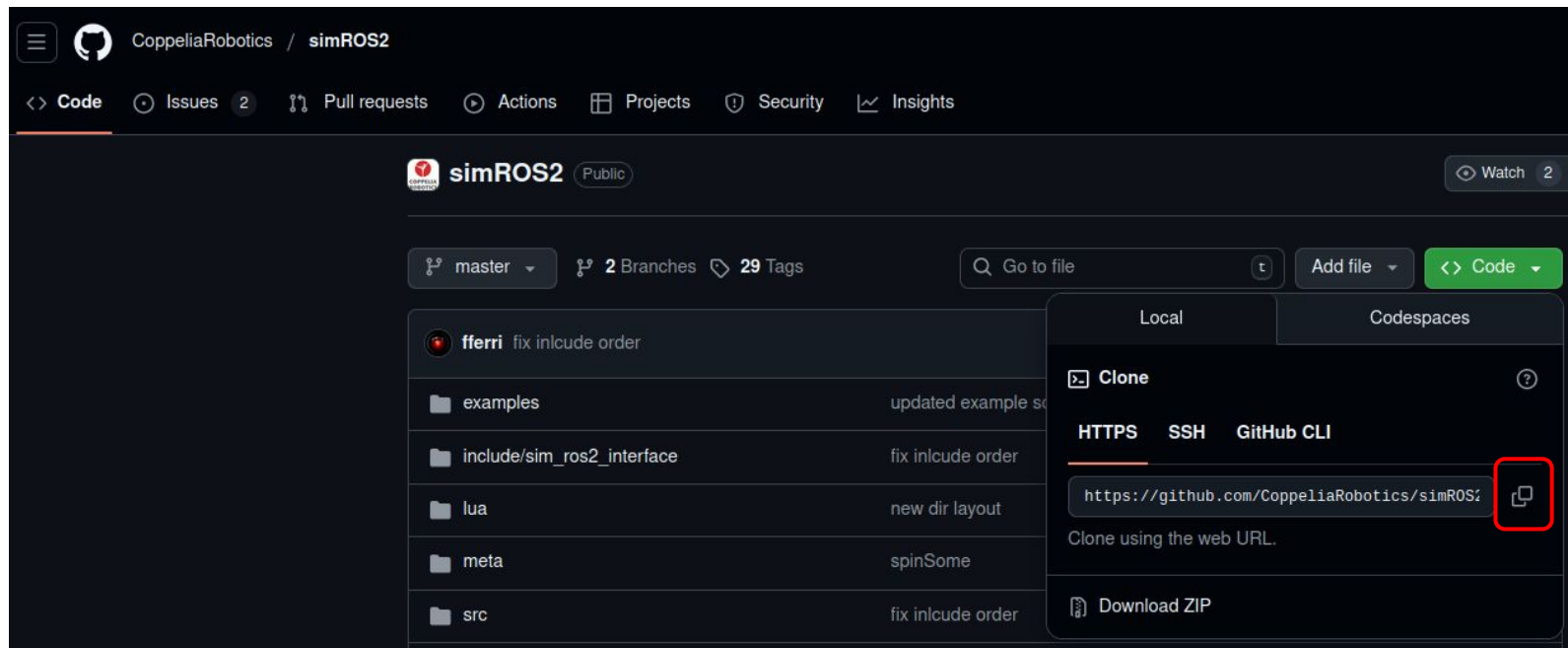
Previously on IA368 II



ROS2

Let's build !

<https://github.com/CoppeliaRobotics/simROS2>



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ffferri fix include order

examples updated example s

include/sim_ros2_interface fix include order

lua new dir layout

meta spinSome

src fix include order

Local Codespaces

Clone ?

HTTPS SSH GitHub CLI

https://github.com/CoppeliaRobotics/simROS2

Clone using the web URL.

Download ZIP

CoppeliaSim + ROS 2

- Although we build `sim_ros2_interface` before using CoppeliaSim with ROS 2, that's not a dependency
- ROS 2 functionality in CoppeliaSim is already supported via the [ROS 2 Interface](#) ([libsimROS2.so](#)) by default
- It was just an example of how to build a cloned package from GitHub

CoppeliaSim + ROS 2 + python/MATLAB/C++

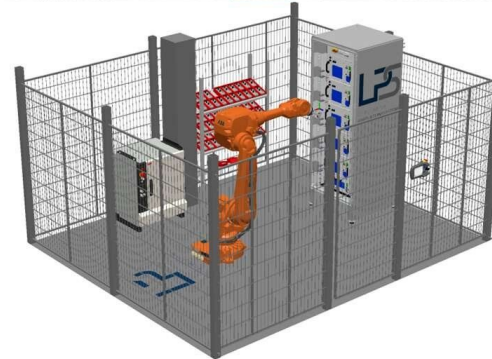
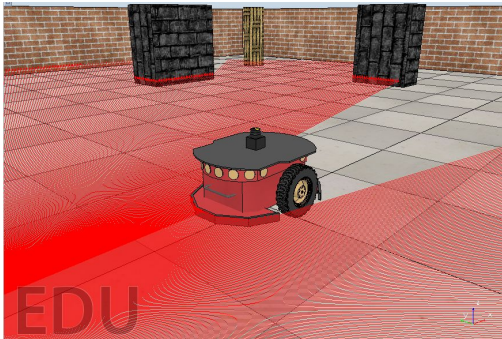


 ROS2

The ROS 2 logo, consisting of a 3x3 grid of dots to the left of the text "ROS2".

CoppeliaSim + ROS 2 + python/MATLAB/C++

- Use in simulation then real life



Real robot+ ROS 2 + python/MATLAB/C++



ROS2



CoppeliaSim + ROS 2 + python/MATLAB/C++

- With all the data being read from the simulator just need to act in the simulation
- Legacy API (Remote API) or ZeroMQ
 - It enable to communicate with the simulator through python
 - For example, the speed for each wheel:
 - “sim.setJointTargetVelocity(jointHandle, vel)”

```
pip install coppeliasim-zmqremoteapi-client
```

ZeroMQ: <https://github.com/CoppeliaRobotics/zmqRemoteApi>



CoppeliaSim + ROS 2 + python/MATLAB/C++

- With all the data being read from the simulator just need to act in the simulation
- Legacy API (Remote API) or ZeroMQ
 - It is possible to create function in the simulator and call in the python:

CoppeliaSIM

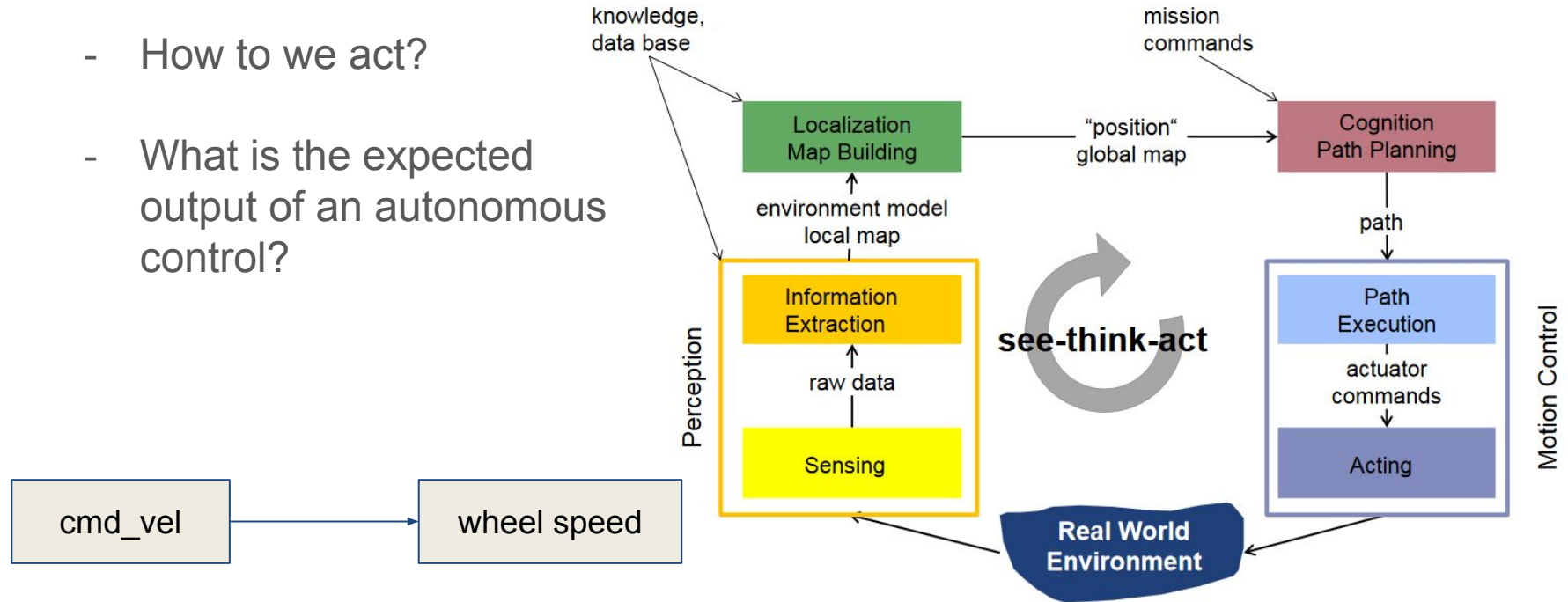
```
function printMessage(inInts, inFloats, inStrings, inBuffer)
    print( inStrings)
end
```

Python

```
self.sim.callScriptFunction(
    "printMessage@./Script",
    self.sim.scripttype_sandbox,
    [], # inputInts (or inputStrings depending on your script)
    [], # inputFloats
    text # inputStrings
)
```

CoppeliaSim + ROS 2 + python/MATLAB/C++

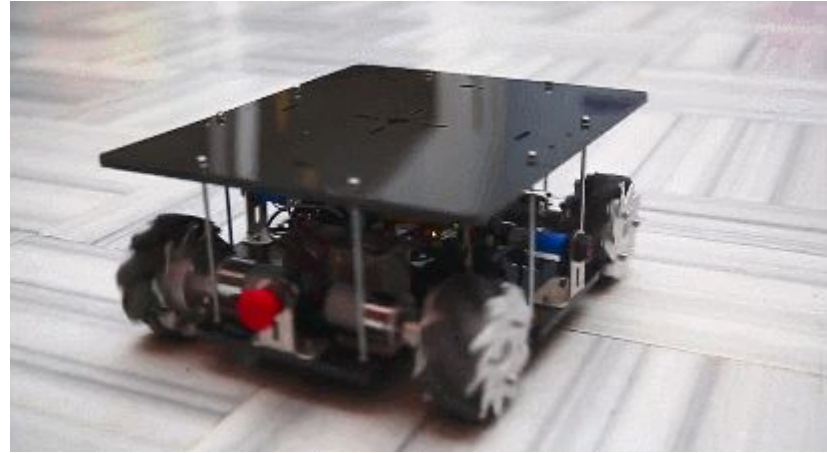
- How to we act?
- What is the expected output of an autonomous control?



Let's Work!

```
from coppeliasim_zmqremoteapi_client import *
```

```
client = RemoteAPIClient()  
sim = client.require('sim')  
sim.loadScene(SCENE_PATH)  
sim.startSimulation()  
.  
.  
.  
sim.stopSimulation()
```



Control your robot through ROS

Hands-On

It's your time to control myRobot with
ROS 2 using our previous classes



 ROS2

Thank You...

...for your attention

